EXPECTED SYZYGIES AND FRÖBERG'S CONJECTURE THROUGH DEGREE $2d$

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Abstract. We prove that the Expected Syzygy Conjecture of Gesmundo, Kayser, and Telen implies Fröberg's predicted Hilbert function through degree $2d$ for general forms of degree d, over every field of characteristic zero and for every number of variables and generators. The argument deforms the degree-d equations of general points. Evaluation of syzygies controls the increase in multiplication rank. At degree $2d$, where the first Koszul relations occur, moving one point shows that every syzygy with zero coefficient evaluations is Koszul whenever the evaluation image has dimension smaller than the number of points.

1. Introduction

Let k be a field of characteristic zero and let $S = k[x_1, \dots, x_n]$, with its standard grading. For general forms $F_1, \dots, F_r \in S_d$, Fröberg's conjecture predicts

$$\mathrm{HS}_{S/(F_1,\dots,F_r)}(u) = \left[\frac{(1-u^d)^r}{(1-u)^n}\right]_+ . \tag{1.1}$$

For a power series with constant coefficient 1, positive truncation $[\sum_{t\geq 0} a_t u^t]_+$ retains the coefficients preceding the first nonpositive coefficient and replaces that coefficient and all later coefficients by zero. This is the truncation used in [1].

The Expected Syzygy Conjecture concerns a different family of ideals: the ideals generated by the lowest-degree equations of general points. Gesmundo, Kayser, and Telen predict the Hilbert function of these *chopped ideals* until it reaches the Hilbert function of the points [2, Conjecture 1]. We show that their conjecture implies the following assertion.

Theorem 1.1. *Assume the Expected Syzygy Conjecture. Let k be a field of characteristic zero, and let $n, d \geq 1$ and $r \geq 0$. For general degree-d forms $F_1, \dots, F_r$ in $S = k[x_1, \dots, x_n]$,*

$$\dim_k(S/(F_1,\dots,F_r))_t = [u^t]\left[\frac{(1-u^d)^r}{(1-u)^n}\right]_+ \qquad (0 \leq t \leq 2d).$$

For each fixed n, d, r, these equalities hold simultaneously on a nonempty Zariski open subset of the coefficient space.

Here and throughout, a statement for *general* forms or points means a statement on a nonempty Zariski open subset of the relevant parameter space. The notation $[u^t]$ denotes coefficient extraction. The conjectural input is stated precisely in Conjecture 2.2 below.

The proof has two parts. Before a chopped ideal reaches the point ideal in a given degree, its conjectural Hilbert function gives a specialization with the required multiplication rank. Once the two ideal components agree, we perturb the generators. Evaluation of syzygy coefficients at the points supplies directions that increase the rank. Below degree $2d$, injective evaluation immediately detects the relevant syzygies. At degree $2d$, the moving-point argument in Lemma 4.2 accounts for the constant Koszul relations. Proposition 5.1 then gives the required Hilbert function at every such target degree.

2. General points and chopped ideals

Write

$$N_j = \dim_k S_j = \binom{n+j-1}{j} \quad (j \geq 0), \qquad N_j = 0 \quad (j < 0),$$

and, for fixed d, r, put

$$c_t = [u^t] \frac{(1-u^d)^r}{(1-u)^n}. \tag{2.1}$$

The coefficients needed in this paper are

$$c_t = N_t - rN_{t-d} \quad (d \leq t < 2d), \qquad c_{2d} = N_{2d} - rN_d + \binom{r}{2}. \tag{2.2}$$

The term $\binom{r}{2}$ records the first Koszul relations.

We carry out the point constructions over an algebraically closed field of characteristic zero. At the end we descend the resulting rank conditions to the original field. For a set $Z = (z_1, \ldots, z_p)$ of distinct points in $\mathbb{P}^{n-1}$, let $I(Z)$ denote its homogeneous vanishing ideal. Choose nonzero representatives of the points so that evaluation has values in k^p; changing the representatives only rescales its coordinates.

Lemma 2.1 (Interpolation and generation)**.** *For $n \geq 2$ and a general set of $p \geq 1$ points, evaluation $S_j \to k^p$ has rank $\min\{N_j, p\}$ in every degree $j \geq 0$. If $p \leq N_{d-1}$ and $J = (I(Z)_d)$, then*

$$J_t = I(Z)_t \qquad (t \geq d). \tag{2.3}$$

Proof. Point evaluations span S_j^*, since a polynomial that vanishes at every projective point is zero. We can therefore choose $\min\{N_j, p\}$ independent evaluations. Maximal rank is an open condition on the irreducible configuration space. Let s be the least integer with $N_s \geq p$ and impose these conditions for $0 \leq j \leq s$. Choose a linear form ℓ nonvanishing at all the points. Multiplication by ℓ acts on evaluation values by an invertible diagonal matrix, so surjectivity in degree s implies surjectivity in every higher degree. This proves the first assertion using finitely many open conditions.

Now suppose evaluation in degree $d-1$ is surjective. Evaluation shows that

$$S_d = I(Z)_d + \ell S_{d-1}.$$

Thus the image of $I(Z)_d$ in $S/(\ell)$ is the whole degree-d component. For $f \in I(Z)_t$ with $t > d$, its image modulo ℓ can consequently be written as a sum of products of images of elements of $I(Z)_d$ and degree-$(t-d)$ forms. Lift this expression to obtain

$$f - \sum_j h_j g_j = \ell f', \qquad g_j \in I(Z)_d, \quad h_j \in S_{t-d}, \quad f' \in S_{t-1}.$$

The left side vanishes at the points, and ℓ does not vanish there, so $f' \in I(Z)_{t-1}$. Induction on t, starting at $t = d$, puts f' and then f in J. This proves (2.3). □

The chopped ideal in degree d is $J = (I(Z)_d)$. Its degree-d component agrees with $I(Z)_d$, but the higher components may initially be smaller. The following is the quotient form of the Expected Syzygy Conjecture of [2, Conjecture 1]; the number of variables here is one greater than the projective dimension.

Conjecture 2.2 (Expected Syzygy Conjecture)**.** *Let $n \geq 2$, let Z be a general set of $p \geq 1$ points in $\mathbb{P}^{n-1}$, and let d be the least nonnegative integer with $N_d \geq p$. Suppose $d \geq 1$. Put $r = N_d - p$, $J = (I(Z)_d)$, and define c_t by (2.1). Let*

$$e_0 = \min\{e \geq 1 : c_{d+e} \leq p\},$$

with $e_0 = \infty$ if the set is empty. Then

$$\dim_k(S/J)_{d+e} = \begin{cases} c_{d+e}, & 0 \leq e < e_0, \\ p, & e \geq e_0. \end{cases} \tag{2.4}$$

The definition requires $e \geq 1$ because $c_d = p$. Subtracting the predicted dimension of the ideal in the original formulation from N_{d+e} gives exactly (2.4). By Lemma 2.1, $\dim_k(S/I(Z))_t = p$ for $t \geq d$. Since $J \subset I(Z)$, the second line of (2.4) therefore gives

$$J_t = I(Z)_t \qquad (t \geq d + e_0). \tag{2.5}$$

We call $d + e_0$ the first plateau degree.

We will also use a lower bound that holds for general forms independently of the conjecture.

Lemma 2.3 (The dimension lower bound)**.** *For $r \leq N_d$ and linearly independent forms $F_1, \ldots, F_r \in S_d$, let $A = S/(F_1, \ldots, F_r)$. Then*

$$\dim_k A_t \geq \max\{c_t, 0\} \qquad (d \leq t \leq 2d). \tag{2.6}$$

Proof. Below degree $2d$, the multiplication map $S_{t-d}^r \to S_t$ has source dimension rN_{t-d}, giving the claimed bound. At degree $2d$, its kernel contains the constant Koszul relations

$$e_i \wedge e_j \longmapsto F_j e_i - F_i e_j \quad \text{from } \bigwedge^2 k^r \text{ to } S_d^r.$$

This map is injective: express each coefficient in a basis of S_d extending $F_1, \ldots, F_r$, and read the coefficient of F_j in component i. The multiplication rank is consequently at most $rN_d - \binom{r}{2}$. Formula (2.2) gives the bound at $2d$. □

3. Evaluation of syzygies and deformation

Fix $n \geq 2$, $d \geq 2$, and $1 \leq r < N_d$, and put $p = N_d - r$. Let Z be p general points and choose a basis

$$Q = I(Z)_d = \operatorname{span}(q_1, \ldots, q_r), \qquad J = (Q).$$

For $t = d + e \leq 2d$, the space of syzygies is

$$\mathcal{Z}_t = \ker\left(S_e^r \longrightarrow S_t\right), \qquad (a_i) \longmapsto \sum_i q_i a_i.$$

Let H_t be the quotient of $\mathcal{Z}_t$ by the degree-t Koszul boundaries. These boundaries are zero when $t < 2d$; at $t = 2d$ they are the constant relations in Lemma 2.3. Every coefficient of a boundary vanishes at Z, so coefficient evaluation induces

$$\varepsilon_t : H_t \longrightarrow (k^r)^p, \qquad [(a_i)] \longmapsto \left((a_i(z_\alpha))_{i=1}^r\right)_{\alpha=1}^p. \tag{3.1}$$

Write

$$G_t = \operatorname{im} \varepsilon_t, \qquad \tau_t = \dim_k G_t, \qquad U_t = \ker \varepsilon_t.$$

These dimensions do not depend on the chosen basis or point representatives. The coordinates of $(k^r)^p$ are grouped into one block for each point.

Lemma 3.1 (Independent evaluation rows)**.** *For any $b \leq \min\{p, \tau_t\}$ distinct point labels, one can choose a scalar functional on each of the corresponding blocks k^r whose pullbacks to H_t are linearly independent.*

Proof. Work on a common open set where the ranks of all block projections of G_t are constant. Permuting the points preserves the construction, up to invertible changes in evaluation coordinates. The generic projection rank onto a set of j blocks therefore depends only on j; call it $f(j)$. These are ranks of sums of row spaces, hence they are submodular. Applying submodularity to two j-element sets intersecting in $j-1$ elements gives

$$2f(j) \geq f(j-1) + f(j+1), \qquad f(0) = 0, \quad f(p) = \tau_t.$$

The successive increments of f are nonincreasing nonnegative integers. Thus

$$f(j) \geq \min\{j, \tau_t\}. \tag{3.2}$$

Indeed, if $f(j) < j$, one of the first j increments is zero; every later increment is then zero, so $f(j) = \tau_t$.

For the chosen blocks, let $V_1, \dots, V_b \subset H_t^*$ be their row spaces. Equation (3.2) says that the sum of any j of them has dimension at least j. This implies that we can choose independent representatives $v_i \in V_i$. Here is an induction proof. If a nonempty proper subcollection has sum dimension equal to its size, first choose representatives there, then apply induction to the other spaces in the quotient by that sum. If there is no such subcollection, choose any nonzero vector in one space and quotient by its span. Every subcollection of the remaining spaces had at least one dimension of slack, so the same condition holds in the quotient and induction applies. Each representative is the pullback of a scalar functional on its block, proving the lemma. □

Proposition 3.2 (Deformation at a target degree)**.** *Suppose $J_t = I(Z)_t$ for some $d \leq t \leq 2d$. Then general degree-d forms $F_1, \dots, F_r$ satisfy*

$$\dim_k (S/(F_1, \dots, F_r))_t \leq \max\{0, p - \tau_t\}. \tag{3.3}$$

Proof. At the tuple $q = (q_1, \dots, q_r)$, the multiplication map

$$\mu_q : S_{t-d}^r \longrightarrow S_t$$

has image $I(Z)_t$ and cokernel k^p, identified by evaluation. Its rank is $N_t - p$. Put $b = \min\{p, \tau_t\}$ and use Lemma 3.1 to choose b independent scalar rows on H_t, supported at b distinct points. Evaluation $S_d \to k^p$ is surjective. We can therefore choose $\delta_1, \dots, \delta_r \in S_d$ whose values realize these rows and are zero at all unselected points.

Set $F_i(u) = q_i + u\delta_i$. The derivative of $\mu_{F(u)}$ from its special kernel to its special cokernel is

$$(a_i) \longmapsto \left(\sum_i \delta_i(z_\alpha) a_i(z_\alpha) \right)_{\alpha=1}^{p}. \tag{3.4}$$

It factors through H_t, since the Koszul coefficients vanish at the points. Its image is the full b-dimensional coordinate subspace on the selected labels, by the independence of the chosen rows. Thus the derivative has rank b.

This first-order rank gives an actual nonzero minor. Choose a nonsingular pivot block of μ_q of size $N_t - p$, and b kernel vectors whose derivatives are independent in the cokernel. After eliminating the pivot block over $k[[u]]$, the additional block is $uB + O(u^2)$ with $\det B \neq 0$. The enlarged minor has a nonzero leading coefficient of order u^b. It follows that the generic multiplication rank is at least $N_t - p + b$. Its cokernel has dimension at most $p - b$, which is (3.3). □

4. The evaluation kernel through degree $2d$

We now determine enough of the evaluation map to apply the deformation bound. In degrees below $2d$, there are no Koszul boundaries, so injective evaluation of coefficient forms is sufficient.

Lemma 4.1. *Suppose* $1 \le e < d$ *and* $J_{d+e} = I(Z)_{d+e}$. *If* $p \ge N_e$, *then*

$$U_{d+e} = 0, \qquad \tau_{d+e} = p - c_{d+e}. \tag{4.1}$$

If $p < N_e$, *general* r *degree-*d *forms generate* S_{d+e} *in that degree, and* $c_{d+e} \le 0$.

Proof. If $p \ge N_e$, evaluation on S_e is injective by Lemma 2.1. A syzygy with zero coefficient evaluations therefore has all coefficients zero. Since the target image is $I(Z)_{d+e}$ and there are no boundaries in this degree,

$$\dim_k H_{d+e} = rN_e - (N_{d+e} - p) = p - c_{d+e}.$$

This proves (4.1).

If $p < N_e$, then $r = N_d - p \ge N_d - N_e + 1$. Choose x_n^d and all degree-d monomials having total degree at least $e+1$ in $x_1, \dots, x_{n-1}$. There are $N_d - N_e + 1$ such generators, since the number with other-variable degree at most e is

$$\sum_{j=0}^{e} \binom{n+j-2}{j} = N_e.$$

Every degree-$(d+e)$ monomial is divisible by x_n^d, or has other-variable degree at least $e+1$ and admits a degree-d divisor of that type. The latter uses $e+1 \le d$. Thus the multiplication map is surjective for this choice and, by openness, for general r generators. Counting its source gives $rN_e \ge N_{d+e}$, hence $c_{d+e} \le 0$. □

At degree $2d$, the Koszul boundary map is injective, as in Lemma 2.3. When $J_{2d} = I(Z)_{2d}$, this gives

$$\dim_k H_{2d} = rN_d - (N_{2d} - p) - \binom{r}{2} = p - c_{2d}. \tag{4.2}$$

The following lemma handles the kernel of coefficient evaluation at this endpoint.

Lemma 4.2 (Moving one point)**.** *In characteristic zero, for general* Z,

$$\tau_{2d} < p \quad \Longrightarrow \quad U_{2d} = 0. \tag{4.3}$$

Consequently, if $J_{2d} = I(Z)_{2d}$, *then*

$$\tau_{2d} = p - c_{2d} \quad \text{if } c_{2d} > 0, \qquad \tau_{2d} \ge p \quad \text{if } c_{2d} \le 0. \tag{4.4}$$

Proof. Suppose first that $p < N_{d-1}$. Choose a nonzero $g \in I(Z)_{d-1}$. The forms $x_1 g, \dots, x_n g$ are independent and can be included in the basis of Q. For each point z_α, surjective evaluation on S_{d-1} gives a separator h_α that vanishes at every other point and is nonzero at z_α. Choose i with $x_i(z_\alpha) \ne 0$ and $j \ne i$. The syzygy

$$h_\alpha(x_i e_{x_j g} - x_j e_{x_i g})$$

has coefficient degree d and a nonzero evaluation supported only at z_α. Here $e_{x_j g}$ denotes the basis label of the generator $x_j g$. The evaluations for the p different points are independent, so $\tau_{2d} \ge p$.

It remains to prove (4.3) when $p \ge N_{d-1}$ and $\tau_{2d} < p$. Represent a class in U_{2d} by a syzygy $a = (a_i)$. Each coefficient lies in $I(Z)_d = Q$, so $a = Mq$ for a scalar matrix M, with q viewed as a column vector, and $q^T M q = 0$. The skew-symmetric part of M represents a Koszul boundary. After subtracting that boundary, we may assume M symmetric.

To differentiate the relation, both the basis q and the relation matrix M must vary with the points. Choose affine charts and a nonzero evaluation minor giving a locally regular basis $q(Z)$ of Q. The map

$$\{\text{symmetric } r \times r \text{ matrices}\} \longrightarrow S_{2d}, \qquad M \longmapsto q(Z)^T M q(Z),$$

has regular entries in the point coordinates. On a nonempty open set its rank is constant, so its kernel is a vector bundle. Every kernel vector extends to a local section. We may therefore extend our matrix to a locally regular symmetric matrix $M(Z)$ satisfying $q(Z)^T M(Z) q(Z) = 0$.

Move only z_α with projective tangent velocity v. Differentiation of this identity gives the syzygy

$$b = \dot{M} q + 2M\dot{q}.$$

At a stationary point z_β, both $q(z_\beta)$ and $\dot{q}(z_\beta)$ vanish. At the moving point, differentiating $q(Z; z_\alpha(Z)) = 0$ gives $\dot{q}(z_\alpha) = -D_x q(z_\alpha) v$. The coefficient evaluation of b is thus supported only at z_α, with value

$$-2MD_x q(z_\alpha) v. \tag{4.5}$$

On a common generic open set, the dimension of G_{2d} and the ranks of its projections away from each single block are constant. The subspaces of G_{2d} supported at individual points then have equal dimensions by permutation symmetry. If that common dimension were positive, their direct sum would have dimension at least p inside G_{2d}, contrary to $\tau_{2d} < p$. Every such subspace is therefore zero. Equation (4.5) vanishes for every point and every projective tangent direction, so $MD_x q(z_\alpha) v = 0$.

Each component f of Mq vanishes at Z and has zero projective differential there. Its radial derivative also vanishes by homogeneity. Hence every homogeneous partial derivative of f vanishes at Z. These derivatives have degree $d-1$, and evaluation in that degree is injective because $p \geq N_{d-1}$. They are zero polynomials. In characteristic zero, a polynomial whose partial derivatives are all zero is constant; since f is homogeneous of positive degree, $f = 0$. Independence of the q_i gives $M = 0$. The original syzygy class is consequently zero, proving (4.3).

Finally assume equality with the point ideal in degree $2d$. If $c_{2d} > 0$, then (4.2) gives $\tau_{2d} \leq p - c_{2d} < p$; the proved implication makes evaluation injective, so $\tau_{2d} = p - c_{2d}$. If $c_{2d} \leq 0$ and $\tau_{2d} < p$, the same implication would give $\tau_{2d} = p - c_{2d} \geq p$, a contradiction. This proves (4.4). □

5. The implication for general forms

Proposition 5.1 (Equality at a target degree)**.** *Let $n, d \geq 2$, $1 \leq r < N_d$, and $p = N_d - r$. For p general points Z, suppose $J = (I(Z)_d)$ satisfies $J_t = I(Z)_t$ for a degree $d \leq t \leq 2d$. Then general degree-d forms satisfy*

$$\dim_k \bigl(S/(F_1, \ldots, F_r)\bigr)_t = \max\{c_t, 0\}. \tag{5.1}$$

Proof. For $t = d$, the assertion is ordinary linear algebra: r general forms are independent, and $\dim_k A_d = N_d - r = p$. For $d < t < 2d$, if $p < N_{t-d}$, Lemma 4.1 gives $A_t = 0$ and $c_t \leq 0$. Otherwise that lemma gives $\tau_t = p - c_t$, and Proposition 3.2 gives $\dim_k A_t \leq \max\{c_t, 0\}$. At $t = 2d$, Lemma 4.2 gives the same upper bound: use $\tau_{2d} = p - c_{2d}$ when $c_{2d} > 0$ and $\tau_{2d} \geq p$ otherwise. In every case the opposite inequality is Lemma 2.3, proving (5.1). □

Proof of Theorem 1.1. We first work over an algebraic closure of k. If $r = 0$, the assertion is the Hilbert series of S. If $n = 1$ and $r \geq 1$, the quotient is $k[x_1]/(x_1^d)$. If $d = 1$, general linear forms give a polynomial ring in $\max\{n - r, 0\}$ variables. If $r \geq N_d$, general forms span S_d and the quotient vanishes from degree d onward.

For $1 \leq r \leq n$, specialize to $x_1^d, \ldots, x_r^d$. Their quotient has Hilbert series

$$\frac{(1 + u + \cdots + u^{d-1})^r}{(1-u)^{n-r}} = \frac{(1-u^d)^r}{(1-u)^n}.$$

Its coefficients are nonnegative and remain zero after any first zero, so this is also its positive truncation. The ranks of the specialized multiplication maps give the upper bound $\dim_k A_t \leq c_t$ for general forms through degree $2d$. Lemma 2.3 gives the reverse bound. Thus the theorem holds in these cases as well.

We may now assume $n, d \geq 2$ and $n < r < N_d$. Put $p = N_d - r$, choose p general points, and let $J = (I(Z)_d)$. If $p \leq N_{d-1}$, Lemma 2.1 gives $J_t = I(Z)_t$ for every $t \geq d$. Proposition 5.1 therefore applies in every degree $d \leq t \leq 2d$.

If $p > N_{d-1}$, the least interpolation degree of the points is d, so Conjecture 2.2 applies. At $t = d$, both the special and general quotients have dimension $p = c_d$. Before the first plateau, the special quotient has dimension $c_t > p$. Specialization can only decrease the rank of multiplication, so a general quotient has dimension at most c_t. Lemma 2.3 gives equality. At and after the plateau, equation (2.5) and Proposition 5.1 apply. The same argument covers $e_0 = \infty$, since then every required degree precedes the plateau.

We have proved

$$\dim_k A_t = \max\{c_t, 0\} \qquad (d \leq t \leq 2d). \tag{5.2}$$

Each equality is a nonempty open rank condition, and the coefficient space of ordered tuples is irreducible. Their finite intersection is nonempty. On that intersection a standard graded quotient that is zero in one degree is zero in every later degree. Consequently the simultaneous equalities (5.2) cannot be followed by a positive raw coefficient within the stated range after a nonpositive one. Together with the unchanged components below d, they give precisely the positive truncation in (1.1) through degree $2d$.

Finally, all multiplication matrices have entries polynomial in the coefficients of the F_i, over the original field. A rank minor nonzero over its algebraic closure is already a nonzero polynomial over that field. The finite collection of required nonvanishing minors therefore defines a nonempty open set over k. Since k has characteristic zero and is infinite, this open set contains k-rational tuples. The lower bounds are valid over k as well, completing the proof. □

References

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